

Q1. Program based on Classes and Objects - Initializing Data member using Objects directly

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
class Student {
    int rollNo;
    String name;
    double marks;
    void displayDetails() {
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Student student = new Student();
        student.rollNo = sc.nextInt();
        sc.nextLine();
        student.name = sc.nextLine();
        student.marks = sc.nextDouble();
        student.displayDetails();
    }
}
```

INPUT

```
192321109
pavendhan a
60
```

OUTPUT

```
Roll No: 192321109
Name: pavendhan a
Marks: 60.0
```

Q2. Classes and Objects-Initialize Object Data Using Setter Methods

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
class Student {
    private int rollNo;
    private String name;
    private double marks;
    public void setRollNo(int rollNo) {
        this.rollNo = rollNo;
    }
    public void setName(String name) {
        this.name = name;
    }
    public void setMarks(double marks) {
        this.marks = marks;
    }
    public void displayDetails() {
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Student student = new Student();
        int rollNo = sc.nextInt();
        sc.nextLine();
        String name = sc.nextLine();
        double marks = sc.nextDouble();
        student.setRollNo(rollNo);
        student.setName(name);
        student.setMarks(marks);
        student.displayDetails();
    }
}
```

Q3. Student Details Using Default and Parameterized Constructors

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;

public class Student {
    int rollNo;
    String name;
    double marks;

    Student() {
        rollNo = 101;
        name = "Rahul";
        marks = 85.5;
    }

    Student(int rollNo, String name, double marks) {
        this.rollNo = rollNo;
        this.name = name;
        this.marks = marks;
    }

    void displayDetails() {
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Student s1 = new Student();
        s1.displayDetails();
        int rollNo = sc.nextInt();
        sc.nextLine();
        String name = sc.nextLine();
        double marks = sc.nextDouble();
        Student s2 = new Student(rollNo, name, marks);
        s2.displayDetails();
    }
}
```

Q4. Net Salary Calculation Using Single Inheritance

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int empId = sc.nextInt();
        sc.nextLine();
        String empName = sc.nextLine();
        double basicSalary = sc.nextDouble();

        Salary s = new Salary();
        s.getEmployeeDetails(empId, empName, basicSalary);
        s.calculateSalary();
        s.displayDetails();

        sc.close();
    }
}

class Employee {
    int empId;
    String empName;
    double basicSalary;

    void getEmployeeDetails(int empId, String empName, double basicSalary) {
        this.empId = empId;
        this.empName = empName;
        this.basicSalary = basicSalary;
    }
}
```

Q5. Employee Gross Salary Calculation Using Multilevel Inheritance

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int empId = sc.nextInt();
        sc.nextLine();
        String empName = sc.nextLine();
        double basicSalary = sc.nextDouble();
        GrossSalary gs = new GrossSalary();
        gs.getEmployeeDetails(empId, empName);
        gs.getBasicSalary(basicSalary);
        gs.calculateSalary();
        gs.displayDetails();

        sc.close();
    }
}

class Employee {
    int empId;
    String empName;

    void getEmployeeDetails(int empId, String empName) {
        this.empId = empId;
        this.empName = empName;
    }
}

class Salary extends Employee {
    double basicSalary;

    void getBasicSalary(double basicSalary) {
        this.basicSalary = basicSalary;
    }
}
```